

Contents

▪ Abstract.....	1
1. Introduction.....	2
2. Literature Review.....	2
3. Related Work.....	2
4. Methodology.....	3
5. Result Analysis.....	3
6. Conclusion.....	4
▪ Reference.....	4

Abstract

This report presents a comprehensive analysis of nutritional data collected through web scraping from the Nutrition Value website. The primary objective of this study was to develop a structured nutritional dataset by automatically extracting information from an online source and to demonstrate the practical application of data science techniques in the field of nutrition analytics. Traditional data collection methods often require significant time and manual effort, whereas web scraping provides an efficient approach for gathering large amounts of information within a short period. Python was used as the main programming language for this project because of its powerful data collection and analysis capabilities. The Requests library was used to access the website, while BeautifulSoup was applied to extract nutritional information from the HTML structure of individual web pages. After the data extraction process was completed, the collected information was processed and organized using the Pandas library. The final dataset was then converted into CSV format for further analysis and future use. The completed dataset contained 1,666 food records and 12 nutritional attributes, including food names, serving sizes, calories, protein, fat, carbohydrates, fiber, sugar, sodium, cholesterol, and potassium values. A preliminary analysis of the dataset revealed considerable variation in the nutritional composition of different food items, highlighting the diversity of dietary information available in the collected data. The findings of this study demonstrate that web scraping is an effective method for transforming unstructured web content into a well-organized dataset. The generated dataset can support future research in nutrition science, healthcare, dietary planning, and machine learning. This project also illustrates the importance of combining web technologies with data science tools to create meaningful and reusable nutritional databases.

1. Introduction

Nutrition plays an important role in maintaining human health and preventing disease. People often need reliable nutritional information to make better dietary decisions, but collecting and organizing such information manually can be time-consuming. Data scraping provides an efficient method for automatically collecting large amounts of information from websites and transforming them into structured datasets. In this project, the Nutrition Value website was selected as the data source because it contains a large collection of food-related nutritional information. The main objective of this study was to collect nutritional data from different food products, organize the information into a dataset, and perform a basic analysis to identify patterns within the collected data.

2. Literature Review

Previous studies have shown that web scraping is an effective technique for collecting publicly available information from online sources. Researchers have used web scraping in many fields, including healthcare, finance, education, and social media analysis. In the healthcare and nutrition sectors, nutritional datasets are commonly used to study food composition, calorie distribution, and dietary patterns. Python has become one of the most popular programming languages for web scraping because of its extensive library support. BeautifulSoup is widely used for extracting information from HTML documents, while Pandas provides powerful tools for data cleaning and analysis. These technologies have significantly reduced the effort required to collect and process large amounts of nutritional information.

3. Related Work

Several nutrition-related projects have focused on collecting food composition data from online databases and nutritional websites. Many studies have analyzed calories, carbohydrates, fats, proteins, and micronutrients to understand dietary trends and evaluate food quality. Similar projects have also used machine learning techniques to classify food items and predict nutritional values. However, this project focused primarily on the data collection process through web scraping and on creating a structured dataset for future analysis. Unlike many previous studies that

relied on pre-existing datasets, this work generated an original dataset directly from an online nutritional database.

4. Methodology

The methodology of this project was designed to collect, organize, and analyze nutritional data through web scraping. The first step was selecting the Nutrition Value website as the data source because it provides a large amount of publicly available nutritional information in a structured format. After selecting the website, Python was used to develop the scraping program. The Requests library was applied to access the web pages, while BeautifulSoup was used to extract the required information from the HTML content. The scraping process focused on collecting important nutritional attributes from each food item. The extracted information included the food name, category, serving size, calories, protein, total fat, carbohydrates, fiber, sugar, sodium, cholesterol, and potassium. The scraper automatically visited multiple web pages and collected data from each available food entry. This approach reduced manual effort and allowed a large amount of information to be gathered in a short period. After the data collection phase, the extracted data was converted into a structured format using the Pandas library. The dataset was then cleaned by removing incomplete records and organizing the columns into a consistent structure. Finally, the processed data was exported to a CSV file containing 1,666 food records and 12 nutritional features. The completed dataset was analyzed to verify data accuracy and identify general nutritional patterns among different food items.

5. Result Analysis

The final dataset contained 1,666 food records and 12 different attributes. The analysis showed that the dataset included a wide variety of food categories with significant differences in nutritional composition. Calories varied considerably among different food products, demonstrating the diversity of energy content within the dataset. Protein values ranged from very low to relatively high levels, while carbohydrates represented one of the most variable nutritional components. The dataset also provided important information about sugar, sodium, and cholesterol levels, which are often associated with dietary health risks. The collected information was successfully organized

into a structured format that can support future statistical analysis, visualization, and machine learning applications.

6. Conclusion

This study successfully demonstrated the use of web scraping techniques for collecting nutritional information from an online source. The combination of Python, BeautifulSoup, and Pandas provided an efficient method for extracting, organizing, and analyzing nutritional data. The resulting dataset offers valuable information that can be used in future research related to healthcare, nutrition, dietary planning, and data science. The project also highlights the importance of automated data collection methods in transforming unstructured web content into meaningful and reusable datasets.

Reference

1. Nutrition Value. Nutrition Facts Database. Retrieved from <https://www.nutritionvalue.org/>
2. Richardson, L. BeautifulSoup Documentation.
3. McKinney, W. Python for Data Analysis. O'Reilly Media.
4. The Pandas Development Team. Pandas Documentation.